

JOY NAOMI OLUSANYA

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EDUCATION

Bachelor of Arts in Linguistics

January 2021 – September 2025

Obafemi Awolowo University, Osun State, Nigeria

Thesis Title: Tone in Yoruba Automatic Speech Recognition: Evaluating the Impact of Tone Recognition on Transformer-Based ASR Models

Manuscript accepted to LoResLM Workshop @ EACL 2026

Project Supervisor: Dr. Kolawole Adeniyi

LINGUISTIC SKILLS

- Linguistics & Computational Linguistics (Phonetics, Phonology, Morphology, Syntax, Semantics, Sociolinguistics)
- Language Documentation, Tone Analysis, Data Validation, Corpus Analysis, Data Annotation, Dialectology
- Research: Literature Review, Experimental Design, Data Collection, Dataset Curation, Statistical Analysis
- Linguistics Software (Praat, ELAN, FLEx, Audacity)
- Language Proficiency: Yoruba, English, Nigerian Pidgin English, German

TOOLS & FRAMEWORKS

• Python (pandas, NumPy, scikit-learn, matplotlib) • Julia • SQL • Deep Learning Frameworks (PyTorch, TensorFlow, Keras, JAX, Scikit-learn) • NLP Language Models (Transformers, Hugging Face, NLTK, spaCy, Machine Translation, ASR, LLM Evaluation) • Model Evaluation (BLEU, ROUGE, BERTScore, COMET, Human Annotation, Benchmarking) • MLOps Tools (Docker, MLflow, W&B, GitHub, AWS, GCP) • Web Frameworks / APIs (Streamlit, FastAPI, Flask, LangChain, OpenAI API, Gemini API, Tavily API)

PUBLICATIONS

The Last Translation Benchmark ([link](#))

Published on arXiv, 2026

Activation Editing for Low-Resource Cross-Lingual Machine Translation: Does JoLA Transfer? ([link](#))

Accepted to the Women in Machine Learning (WiML) Symposium @ ICML 2026, International Conference on Machine Learning, 2026

Tone in Yoruba Automatic Speech Recognition: Evaluating the Impact of Tone Recognition on Transformer-Based ASR Models ([link](#))

Accepted to LoResLM Workshop @ EACL 2026, European Chapter of the Association for Computational Linguistics, 2026

ÒWE-YOR: Leveraging Transformer-Based Models for Yoruba Proverb Classification ([link](#))

Accepted to WiML @ NeurIPS 2025, Women in Machine Learning Workshop, 2025

WORK EXPERIENCE

Tonative

NLP Researcher & Yoruba Language Lead

Lagos, Nigeria

August 2025 – September 2026

- Developed multilingual NLP datasets for low-resource African languages, ensuring linguistic diversity and cultural relevance.
- Contributed to machine translation research by conducting a machine translation of English into Kikuyu, Kinyarwanda, and Yoruba for multilingual NLP systems.
- Directed language data collection, annotation, validation, and benchmarking workflows to improve model quality and performance.
- Collaborated on multilingual NLP system design to enhance accessibility, inclusivity, and translation quality across African languages.
- Validated English-to-Yorùbá machine translation datasets to improve linguistic accuracy, translation quality, and cultural representation.
- Certification of Contribution: [View Certificate](#)

FriendnPal

AI Developer

Lagos, Nigeria

January 2023 – March 2025

- Developed NLP-powered mental health chatbot systems to improve conversational relevance, contextual understanding, and user engagement.
- Implemented Retrieval-Augmented Generation (RAG) pipelines, preprocessing workflows, and multilingual data systems for scalable AI applications.
- Curated and analysed multilingual datasets to improve model reliability, response quality, and evaluation performance.

Atom Group (Contract)

AI/LLM Developer

Lagos, Nigeria

April 2025

- Developed a commercial AI agent with Instagram scraping pipelines for social media intelligence and automation.
- Built modular API integration and automated data collection systems to improve efficiency and scalability.
- Improved scraping accuracy and reduced system failures through reusable utilities and exception handling.

Brilla AI Project, Kwame AI Ghana

AI Developer

Remote

January 2024 – March 2025

- Developed multilingual speech-to-text and ASR models with accent robustness using deep learning techniques.
- Validated multilingual datasets to improve transcription quality and model reliability.
- Built reproducible preprocessing and evaluation workflows for scalable multilingual speech systems.

Rendo AI Nigeria

ML/NLP Engineer

Lagos, Nigeria

June 2024 – December 2024

- Designed Python-based web scraping pipelines using Requests, Selenium, and BeautifulSoup.
- Automated large-scale data acquisition workflows, improving dataset completeness and operational efficiency.
- Developed reusable scraping and preprocessing tools for structured multilingual datasets.

Cohere AI

Open Source Contributor

Remote

July 2024 – August 2024

- Validated 500+ multilingual translations for AI benchmarks, ensuring linguistic accuracy, consistency, and quality assurance.
- Collaborated with international open-source contributors on multilingual NLP evaluation and benchmarking tasks.
- Certification of Contribution: [View Certificate](#)

LEADERSHIP EXPERIENCE

Center for Low-Resource Languages and Cultures (CLRLC)

Founder & CEO

Nigeria

2025 – Present

- Established a research and advocacy center dedicated to NLP research, language and cultural documentation, and the preservation of low-resource languages and cultures in AI.
- Organised the [CLRLC-LLMs Workshop @ NeurIPS 2025](#), bringing together researchers, practitioners, and community partners working on low-resource language technologies.
- Leading initiatives focused on multilingual NLP, culturally aware AI systems, and community-centred research collaborations.

CLRLC-LLMs Workshop @ NeurIPS 2025

Workshop Chair

Mexico City, Mexico

September 2025

- Established an international research workshop accepted at NeurIPS 2025 focused on advancing low-resource language technologies and cultural representation in large language models (LLMs).
- Coordinated the program committee, invited speakers, and interdisciplinary collaborations across academia and industry.
- Designed the technical workshop structure, including speaker sessions, lightning talks, and community engagement activities.

Tonative

Training Manager

Lagos, Nigeria

January 2026 – September 2026

- Designed and implemented training curriculum for participants across diverse linguistic and cultural contexts.
- Reviewed and validated tutors' teaching materials to ensure curriculum alignment and program quality.
- Coordinated learner feedback and performance assessments to improve training effectiveness and engagement.
- Conducted training sessions and mentorship in AI, NLP, and low-resource language technologies. [View Teaching](#)

Data Science Nigeria OAU

Deep Learning / AI Co-Lead

Remote

January 2025 – November 2025

- Tutored and guided team members in deep learning concepts and model development, delivering hands-on sessions on neural networks, CNNs, and model evaluation that empowered 20+ participants to deploy deep learning models.
- Contributed to planning and executing national deep learning events and hackathons.

- Supported bootcamp management and mentorship programs for emerging AI talents.

Women in Data Science (WiDS)

AI / Machine Learning Lead

Ile-Ife, Nigeria

October 2024 – November 2025

- Led AI/ML workshops for women and coordinated technical learning sessions.
- Designed structured learning paths and created educational resources for all skill levels.
- Organized and moderated speaker sessions promoting inclusivity and mentorship in AI.

PROJECTS

Evaluating LLM Performance on Nigerian Pidgin Sentiment

Researcher / NLP Engineer

Remote

2026

- Benchmarked GPT-4o-mini on the NaijaSenti corpus for Nigerian Pidgin sentiment classification.
- Designed zero-shot prompts informed by Pidgin linguistics knowledge.
- Built evaluation pipelines with visualisations and performance analysis.

Yorùbá Proverb Classification

Researcher / NLP Engineer

Remote

July 2024 – September 2024

- Investigated semantic classification of culturally grounded Yorùbá proverbs.
- Compared Naive Bayes, BERT, and AfroLM models for proverb interpretation.
- Conducted morpho-semantic analysis and structured error evaluation.

Language Documentation: Syllable Structure and Phonotactics of Arogbo-Ijaw, Nigeria

Research Assistant, Obafemi Awolowo University

Ondo, Nigeria

June 2024

- Conducted fieldwork to describe Arogbo-Ijaw syllable patterns using ELAN and Praat.
- Developed annotated language resources for preservation and NLP applications.

English–Portuguese NMT

Researcher / Developer

Remote

June 2025

- Fine-tuned Helsinki MarianMT transformer for English–Portuguese machine translation.
- Evaluated translation quality using BLEU benchmarking metrics.

Lexicography Project (EN–Yorùbá)

Compiler / Research Assistant

Obafemi Awolowo University

January 2024 – April 2025

- Compiled a bilingual ecology and climate dictionary for English and Yorùbá.
- Conducted lexical documentation and terminology research.

Automatic Essay Scoring (AES)

Machine Learning Engineer

Remote

June 2025

- Fine-tuned DistilBERT for automated essay scoring tasks.
- Evaluated model performance using RMSE and Pearson correlation.

Speech Emotion Recognition

Deep Learning Engineer

Remote

January 2024 – March 2024

- Built an LSTM-based speech emotion recognition system using MFCC and chroma features.
- Deployed an interactive Streamlit application for inference and testing.

TEACHING & TALKS

- [Introduction to Deep Learning](#) DSN OAU Chapter
- [Fundamentals of Neural Networks](#) DSN OAU Chapter
- [Fundamentals of Artificial Intelligence & Natural Language Processing](#) Tonative Data Academy
- [Introduction to Data & Types of NLP Datasets](#) Tonative Data Academy
- [An Introspective Look at the African AI Landscape](#) Panelist, Deep Learning Indaba 2026

PROFESSIONAL DEVELOPMENT

[Advanced Language Processing School \(ALPS 2026\) – Accepted Participant](#) Centre Paul Langevin, French Alps
[View more details](#) March 29 – April 3, 2026

[Deep Learning Indaba 2026 – Workshop Organiser](#) Pan-Atlantic University, Lagos, Nigeria
[View more details](#) August 2 – 7, 2026

Deep Learning IndabaX Nigeria 2026 – Attendee

[View more details](#)

University of Ibadan, Nigeria

May 11 – 14, 2026

Deep Learning Indaba 2025 – Attendee

[View more details](#)

Kigali, Rwanda

August 17 – 22, 2025

Data Science Nigeria AI Bootcamp 2025 – Participant

[View more details](#)

Lagos, Nigeria

October 20 – 24, 2025

PEER REVIEWER FOR CONFERENCE/JOURNALS

- **Deep Learning Indaba 2026** Poster Judge (In-person) 2026
- **NeurIPS 2026 Main Conference & Ethics** Reviewer 2026
- **Deep Learning Indaba Awards 2026** Reviewer 2026
- **Deep Learning Indaba 2026 Startup & Dataset Track** Reviewer 2026
- **Deep Learning Indaba Ideathon** Reviewer 2025
- **ICML 2026 Pluralistic Alignment Workshop** Reviewer 2026
- **Tonative × YPIT Storytelling Hackathon** Reviewer 2026

AWARDS & GRANTS

- **NeurIPS 2025 Scholar Award** 2025
- **WiML @ NeurIPS 2025 Travel Funding Award** 2025
- **Deep Learning Indaba 2025 Travel Grant** 2025
- **EACL 2026 Registration Grant** 2026
- **ACL Membership Registration Grant** 2026
- **PyCon Africa 2026 Travel & Registration Grant** 2026

MENTORING

NeurIPS 2026 Affinity Group Mentorship

Mentor

Remote

2026 – Present

- Guided mentees in research development and preparation of research paper submissions for NeurIPS 2026 workshops.
- **Mentees:** Stacy Cheruto; Yousra Farhani.

Deep Learning Indaba 2026

Mentor

Lagos, Nigeria

2026

- Provided in-person mentorship and guidance to emerging researchers and AI practitioners during the conference.

LevelUp Techies Bootcamp

AI Mentor

Remote

September 2024 – November 2024

- Mentored over 50 participants in Python programming and machine learning fundamentals.